Discover Artificial Intelligence

RESEARCH Open Access

Emotion analysis and management improvement of hotel user online reviews based on BERT-RF and PLS-SEM integration

Xiangwei Xia[1*], Chao Qu[1] and Yi Qu[1]

*Correspondence:
Xiangwei Xia
xiaxiangwei2024@126.com
[1]Department of Hospitality Management, Hospitality Institute of Sanya, Sanya 572000, China

Abstract

Online reviews have become a core factor affecting hotel booking decisions and brand reputation. Existing research mostly focuses on emotion analysis techniques, but lacks guidance on the implementation of management strategies, and traditional single text analysis has limitations. Therefore, to accurately locate hotel service issues, reduce the impact of negative reviews, enhance customer trust and platform ratings, an emotion analysis framework and real-time feedback mechanism that integrates multi-source text features are proposed. This study combines Bidirectional Encoder Representations from Transformers with Random Forest (BERT-RF) for emotion analysis, and uses structural equation modeling to model satisfaction and calculate satisfaction index. The emotion classification accuracy of the BERT-RF model reached 90%, with a positive review recall rate of 93% and an F1 score of 91%. The value perception had the strongest impact on satisfaction, with customer satisfaction directly driving loyalty and customer complaints negatively affecting value perception. The study accurately identifies negative reviews through multi-modal emotion analysis, and combines PLS-SEM quantitative satisfaction driving mechanism to provide data support for optimizing service resource allocation in hotels, promoting the transformation from "passive response to negative reviews" to "active prevention" management paradigm. Unlike conventional hybrid models that merely aggregate features, the proposed BERT-RF framework introduces a structured fusion mechanism. The deep context embedding of BERT is dynamically weighted and enhanced based on random forest statistics and domain-specific features, synergistically enhancing the robustness of sentiment classification, especially for fuzzy and domain-specific expressions in hotel reviews.

Keywords Partial least-square method, Structural equation modeling, Hotel management, Satisfaction, Emotion analysis

1 Introduction

With the vigorous development of e-commerce, online reviews have deeply penetrated into the consumer decision-making chain and become the key driving factor influencing hotel booking behavior and shaping brand reputation [1]. Nearly 90% of potential guests

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

will repeatedly review the reviews of historical guests before confirming their booking, viewing them as a touchstone for measuring the hotel's true service level [2]. More intuitively, for every one point increase in a hotel's overall rating on mainstream booking platforms, there is an average potential premium space of up to 11.2% for its room rate [3]. This fully demonstrates that positive word-of-mouth evaluation not only symbolizes reputation, but also directly translates into core competitiveness of economic benefits. An influential negative review with the potential to cause a loss of up to 10% in hotel revenue [4]. It may not only scare off potential customers, but also damage brand image and long-term customer loyalty. Therefore, accurately analyzing the emotion of reviews, especially quickly identifying the key issues reflected in negative reviews, and making targeted and systematic improvements is of great significance for improving customer satisfaction, repeat purchase rates, and establishing strong brand loyalty [5, 6]. Existing research has mostly focused on emotion analysis techniques, but lacks guidance on the implementation of hotel management strategies, and some hotels rely on short-term means such as "brushing positive reviews" to truly improve service quality. Therefore, the study proposes a Partial Least-Square Method-Structural Equation Modeling (PLS-SEM) using bidirectional encoders and Random Forest (RF), aiming to help hotels accurately locate problems, reduce the impact of negative reviews, improve customer trust, and improve online platform ratings.

The key contribution of this research lies in explicitly distinguishing methodological integration from substantive innovation.

(1) For BERT-RF, the innovation lies in a "Feature Importance-Guided Re-weighting" strategy. Rather than simple feature concatenation, RF is employed during training to evaluate the incremental information gain contributed by statistical features relative to BERT's contextual embeddings. Features with low gain are down-weighted, while domain-specific cues are amplified, thereby adaptively bridging the gap between deep semantics and surface statistics, a common challenge in short review texts.
(2) For the PLS-SEM component, the improvement is not algorithmic but architectural and applicational: the CCSI model is extended with hotel-specific constructs. More importantly, the quantitative sentiment scores derived from BERT-RF are used as formative indicators for "Value Perception" and "Quality Perception." This transforms PLS-SEM from a survey-only tool into a hybrid system that directly ingests user-generated content, enabling more ecologically valid and real-time responsive satisfaction analysis.

2 Related works

Online reviews have become a key influencing factor for consumers' hotel choices. Therefore, optimizing the user review management system is crucial for maintaining hotel reputation and market competition. Hidayati used multi-source data analysis methods, word cloud, topic modeling, regression analysis, and PLS-SEM models, to address the single data source generalization in hotel and tourism research. The results indicated significant differences in comment topics, emotion ratings, and length among different platforms, suggesting that relying on a single data source may lead to bias [7]. Li et al. proposed an integrated analysis model that combined econometrics and image processing techniques to address the unclear role of user generated photos in hotel reviews. Comments with user generated images were more helpful, providing a new perspective

and practical inspiration for understanding and managing user generated content [8]. Xu et al. combined bidirectional encoder and long short-term memory network to address the ambiguity, insufficient utilization of contextual semantics, and inadequate quantification of emotion intensity in fine-grained emotion analysis. The method achieved the optimal F1 score in multiple datasets such as hotels and food delivery, and the emotional intensity was quantified. The ablation experiment also verified the effectiveness of each component of the model [9]. Xun et al. proposed a multi-source data analysis method, combining word cloud, topic modeling, regression analysis, and PLS-SEM model, to address the potential bias caused by relying on a single data source in hotel and tourism research. The results showed significant differences in comment topics, emotional tendencies, and length among different platforms, verifying the risk of generalization from a single data source [10]. Kim et al. combined empirical and experimental research on how cross-cultural differences affect online customer comment generation. The results indicated that individualistic cultural users were more inclined to comment based on recent ratings, while collectivist cultural users were more concerned with overall ratings, revealing that differences in thinking patterns were a key driving factor for cross-cultural differences [11].

Wąsowicz et al. proposed a content analysis method that combined quantitative and qualitative methods to assess whether traditional scales accurately reflect consumers' true emotions in hotel service quality evaluation. The model should be adjusted to better fit customer feedback and a new service quality measurement tool was proposed [12]. Petrescu et al. proposed a two-stage mixed research method on how consumers evaluate the online reputation of luxury hotel brands and respond to review fraud behavior, combining interpersonal deception theory and social proof theory. Consumers mainly relied on social proof heuristics and interpersonal communication heuristics to judge the usefulness of reviews, providing theoretical support and practical guidance for brand online reputation management [13]. Lopes et al. proposed a machine learning classification combined with regression analysis method to address the unclear impact of hotel online review response strategies on bookings. The results indicated that strategies such as guiding reviewers to private channels and taking defensive measures could significantly improve bookings. Apologies and informal expressions had negative impacts, providing data support and practical guidance for hotels to optimize their online response strategies [14]. Guo et al. proposed a two-level empirical analysis method, combining negative binomial regression and Tobit regression models, to address the impact of online review information overload on selection decisions. There were differences in the complexity of reviews among hotels, and consumers relied on different information processing paths, providing empirical evidence for hotel managers to identify high-value review features and optimize word-of-mouth management strategies [15]. Kazakov et al. proposed a comprehensive emotion analysis, logistic regression, and semantic network analysis method to address the lack of attention to service elements that were small but affected satisfaction in hotel services. This method identified problems and improved services, enhancing overall satisfaction and providing effective improvement strategies for hotel and destination management [16].

In summary, in the field of emotion analysis and management of hotel user online reviews, scholars have addressed the generalization from a single data source. Some scholars have adopted multi-source data analysis, and teams have proposed integrated

analysis models to clarify the role of user generated photos. However, existing research mainly focuses on the method application and specific problems, and there is still insufficient research on the deep integration of emotion analysis into business scenarios and multi-modal data. Therefore, the study employs a BERT-RF model for emotion analysis of hotel reviews, leveraging the complementary strengths of deep semantic encoding and statistical feature extraction. The PLS-SEM is used to analyze customer satisfaction, thereby supporting hotel review management improvement.

3 Methods and materials

3.1 Emotion analysis method for hotel reviews based on BERT-RF

To address the limitations of existing sentiment analysis methods in handling implicit expressions and domain-specific terminology in hotel reviews, an enhanced BERT-RF fusion framework is proposed. Unlike typical ensemble approaches that merely concatenate features, the proposed method implements a hierarchical feature integration strategy: BERT-generated embeddings serve as the semantic backbone, while RF performs a weighted fusion of complementary statistical and lexical features, including TF-IDF, Word2Vec vectors, and domain emotion dictionary scores, based on their contribution to classification accuracy. This design ensures that the model leverages both deep language understanding and discriminative feature selection, thereby improving both performance and generalizability in hotel review sentiment analysis. In the hotel operation, factors that affect customer satisfaction are complex and diverse, covering multiple dimensions such as service, facilities, and perceived value. The core factors include service quality, facilities and equipment, and perceived value. The secondary factors include hygiene and cleanliness, catering services, convenience and location, and environment and atmosphere. Special influencing factors include disease prevention and control measures and price sensitivity. Customer factors include the gap between expectations and reality, personal preferences and needs, and emotions and scenarios. Under different primary influencing factors, there are also different secondary influencing factors [17]. To assist hotel management personnel in finding improvement and optimization solutions for hotel management from customer review results, this study conducts emotion analysis on hotel reviews to quantify the impact of customer reviews. BERT is a pre-trained language model with bidirectional Transformer encoders, which is stacked with multiple layers of Transformer encoders and capture contextual relationships of words in text through self-attention mechanisms. This model integrates Token Embedding, Segment Embedding, and Position Embedding when converting text into vectors. The key components include CLS tags and SEP tags, where CLS tags are added at the beginning of sentences to represent the entire text, and SEP tags separate different sentences or paragraphs to understand text structure [18]. Figure 1 displays the BERT.

In Fig. 1, the BERT has a three-layer structure, with the top layer being the output layer, the middle layer being the Transformer layer, and the bottom layer being Token Embedding. The pre-training tasks of the BERT include masking the language model and predicting the next sentence. The masked language model randomly masks certain words in a text to predict the masked words based on context. The next sentence prediction is to determine whether two sentences are consecutive paragraphs and train the model to understand the semantic coherence. The model needs to obtain data before training. The study utilizes web crawlers to obtain relevant review data for a certain hotel, including

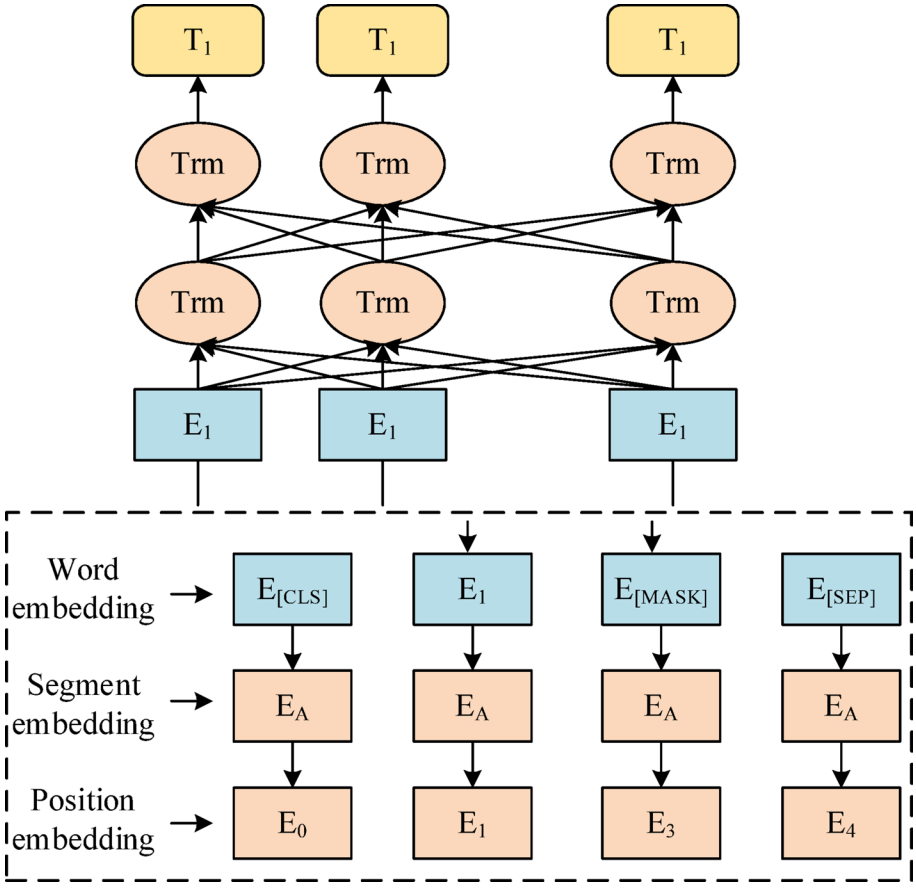


Fig. 1 The structure of BERT

customer booking time, room type, review content, and ratings [19, 20]. To ensure the representativeness and quality of the data, a self-built dataset is constructed with a total of 42,815 online reviews. The reviews come from 32 hotels on China's three major platforms: Ctrip, Meituan and Fliggy. The data collection period covered from January 2023 to December 2024, ensuring temporal relevance and market contemporaneity. Prior to analysis, a comprehensive data cleaning pipeline was implemented: (1) Delete duplicate reviews posted by identical user IDs within a 24-h window; (2) Filter non-textual content (e.g., pure emoticons, ratings without text) and non-Chinese reviews; (3) Eliminate suspected promotional or "brushed" reviews based on excessively repetitive wording and abnormal posting frequency patterns; (4) Exclude comments that are officially flagged by the platform as spam or fake. However, simply mapping numerical ratings to sentiment labels suffers from the well-known "rating-text inconsistency" problem. To ensure label reliability, a three-step annotation protocol is designed: (1) Initial mapping—ratings 1–2 as negative (− 1), 3 as neutral (0), 4–5 as positive (+ 1); (2) Conflict detection—for each review, a text-based sentiment score is computed using the domain lexicon (Eq. 3). If the rating-based label and the lexicon score exhibit strong opposite polarities, the sample is flagged as a boundary case; (3) Manual arbitration—all boundary cases are independently reviewed by two domain experts (experienced hotel managers), who re-label based on the textual content, ignoring the numerical rating. Cases with unresolved disagreement are excluded. Inter-annotator agreement (Cohen's Kappa) reached 0.87. After this procedure, 4.3% of raw samples were re-labeled and 1.2% were excluded, ensuring

that final labels reflect genuine textual sentiment. Based on the above annotation protocol, the finalized sentiment classification tag is determined through a combination of initial mapping and manual arbitration. The baseline mapping rule is expressed as Eq. (1), which serves only as the initial step before conflict detection and human correction.

$$Tag = \begin{cases} -1, & if\ 1 \leq score \leq 2.5 \\ 0, & if\ 2.5 < score \leq 3.5 \\ 1, & if\ 3.5 < score \leq 5 \end{cases} \tag{1}$$

In Eq. (1), Tag represents the customer comment classification tag. *score* represents customer rating. As emphasized in the three-step protocol, this mapping is not the final label but the starting point. All samples that pass the conflict detection and manual arbitration receive their final labels accordingly. After the three-step annotation protocol and manual arbitration, the initial label distribution is heavily skewed toward positive reviews (approx. 68% positive, 20% neutral, 12% negative). To prevent the model from learning a biased decision boundary that favors the majority class, a stratified random down-sampling strategy is applied within each final label category. Before balancing, the raw label distribution (after annotation) is approximately 68% positive, 20% neutral, and 12% negative. After down-sampling, the balanced training set contains approximately 33% samples per category (about 12,000 reviews per class, totaling about 36,000 valid samples). During model training, the training, validation, and test sets are divided in 7:2:1 based on this balanced dataset. The test set comprises approximately 3,600 samples with equal class proportions, ensuring that the reported accuracy (90%) reflects the model's genuine discriminative power across all sentiment classes, rather than a trivial dominance of the positive class [21, 22].

When using the BERT model for emotion analysis of comments, it is necessary to fine tune the BERT. The study chooses the BERT as the base model. When inputting the model, CLS tags are added at the beginning of the sentence to generate the overall text representation. SEP markers are added at the end of the sentence to separate paragraphs or sentences. When constructing the embedding layer, character embedding maps each Chinese character into a vector, segment embedding distinguishes different paragraphs, and position embedding identifies the position information of words in the sequence. Model adjustment includes modifications to the output layer and regularization strategies. The output layer modification converts the output vector marked with CLS into a 2 × 1 dimension, with the first element containing classification task information and the second element serving as the classification output. The regularization strategy involves adding dropout layers and adjusting the final classification output through fully connected testing. To further enhance the accuracy of the BERT model in text emotion classification processing, the study takes RF for feature extraction, which is mainly used for classification and regression tasks. The principle of RF is shown in Fig. 2.

In Fig. 2, RF improves the accuracy of hotel review emotion classification by extracting multidimensional text features and integrating them with the BERT model. When extracting the features of comment text, the number of Chinese characters in the comment text is extracted through regular expressions as the basic feature. The Word2Vec model converts segmented words into vectors. The Gensim module loads pre-trained models. Each word is mapped to a 100-dimensional vector, and all word vectors of a single comment are averaged to generate a comment level vector representation, capturing

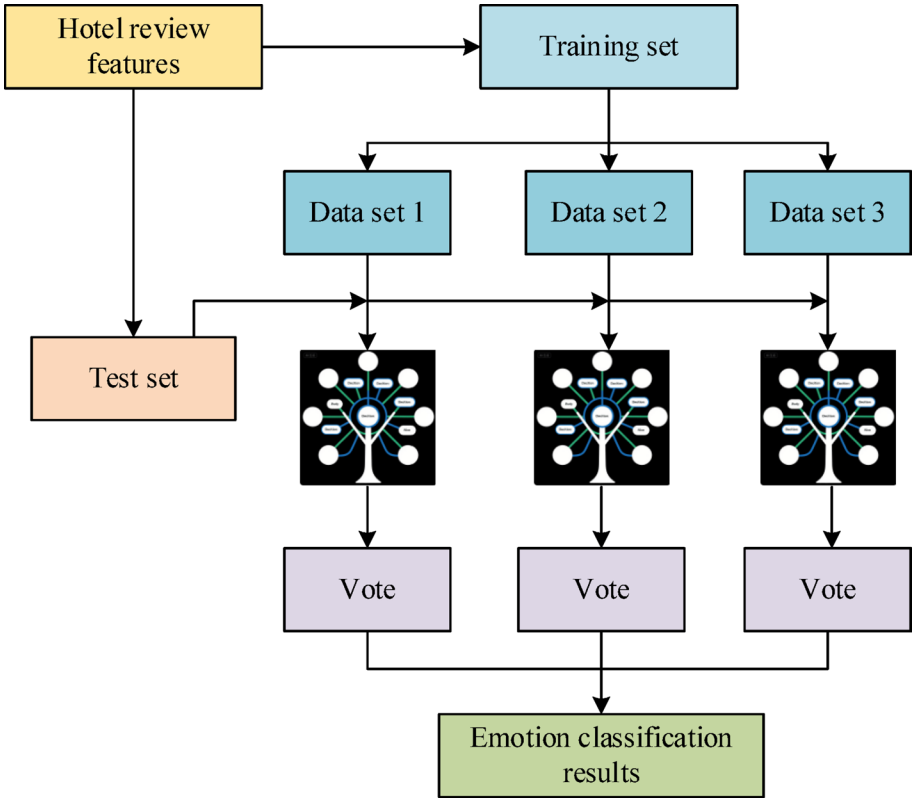


Fig. 2 Principle of random forest text feature extraction

semantic relevance [23, 24]. In addition to this feature, the study also adds Term Frequency-Inverse Document Frequency (TF-IDF) as a supplementary feature. The TF-IDF is presented in Eq. (2).

$$tf-idf\,(words)=\frac{\sum_{i=1}^{W} tf-idf\,(w_i)}{5} \tag{2}$$

In Eq. (2), $tf-idf\,(words)$ represents the average of the top five TF-IDF ranked terms in each comment. W represents the number of terms in the top five TF-IDF rankings in each comment. $tf-idf\,(w_i)$ represents the TF-IDF value of the -th term, where ranges from 1 to 5. After extracting features from hotel reviews, the emotion tendency of the reviews is calculated based on a general emotion dictionary and a hotel domain dictionary. The calculation process is shown in Fig. 3.

The general dictionary contains 24,362 positive words and 28,454 negative words, with an emotional intensity value of ± 1. The domain dictionary contains 187 positive domain terms and 142 negative domain words. Based on the constructed emotion dictionary, the emotion tendency of comments is calculated through dictionary matching and rules. Firstly, the comments are split into multiple short sentence units according to punctuation marks. Next, term matching is performed by traversing the term in each short sentence, matching emotion terms, degree adverbs, and negation words in the emotion dictionary. Finally, weight calculation is performed, assigning corresponding weights based on term types, multiplying and accumulating to obtain the emotion score of the short sentence, and then summarizing it into the overall comment score [25]. The customer comment emotion score is shown in Eq. (3).

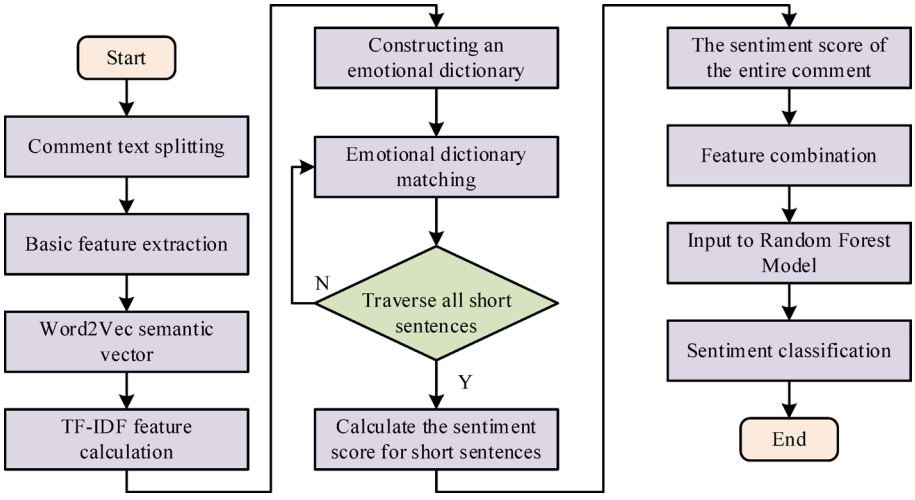


Fig. 3 Comment score calculation process

$$S = \sum_{j=1}^{N} \sum_{i=1}^{m} (-1)^n \times \omega_a \times \omega_b \times \omega_c \tag{3}$$

In Eq. (3), N represents the number of short sentence units in the comment. ω_c signifies the negative term intensity value. m represents the number of emotional terms in a single short sentence. ω_a represents the intensity value of emotional words. n signifies the number of occurrences of negative words in a short sentence. ω_b represents the strength value of degree adverbs. The word count of comments, Word2Vec vector, TF-IDF average, and emotion dictionary score are taken as combined features, and input them into the RF model. To describe the BERT-RF fusion mechanism, the feature construction, concatenation order, and training paradigm are specified as follows:

(1) BERT embedding extraction (frozen): The pre-trained BERT-base-Chinese model (12 layers, 768 hidden units) is used as a fixed feature extractor. For each review, the tokenized text (padded/truncated to 128 tokens) is fed into BERT, and the pooled output of the [CLS] token is extracted, yielding a 768-dimensional vector $\mathbf{v}_{bert}$. Crucially, the BERT parameters are frozen during RF training, that is, no back-propagation flows from RF to BERT. This design choice deliberately decouples deep semantic representation from statistical feature learning, preventing overfitting on the limited hotel review dataset and ensuring that the statistical features (which capture surface-level domain cues) are not suppressed by dominant BERT activations.

(2) Statistical feature construction: In parallel, a 106-dimensional statistical feature vector $\mathbf{v}_{stat}$ is constructed by concatenating:

Word2Vec average (100 dims): Each segmented word is mapped to a 100-dim pre-trained vector, and averaged across all words in the review.
TF-IDF top-5 mean (5 dims): The average of the top-5 TF-IDF values per review (as defined in Eq. 2).
Domain lexicon score (1 dim): The final sentence-level sentiment score (as defined in Eq. 3).

(3) Final input vector and RF training: $\mathbf{v}_{bert}$ and $\mathbf{v}_{stat}$ are concatenated into a single 874-dimensional vector, which serve as the input feature for each review. The RF (with 200 trees, Gini impurity as split criterion) is then trained on this combined feature set using the manually-arbitrated and balanced labels described in the previous annotation protocol. Since RF inherently performs feature importance ranking during training, this architecture allows the model to automatically learn the relative contribution of semantic versus statistical dimensions without requiring joint gradient updates.

The feature extraction of RF can capture statistical features such as word frequency and emotion intensity, which complement BERT's contextual semantic understanding and improve the emotion classification accuracy.

3.2 Hotel customer satisfaction analysis method based on PLS-SEM

After taking the BERT-RF to analyze the emotional tendency of hotel customer reviews towards the hotel, the influencing factors of customer satisfaction are analyzed to enhance the hotel management level in a targeted manner. When analyzing hotel customer satisfaction, the study uses the CCSI as the benchmark model for analysis. This model is an indicator system developed and widely applied in China for systematically measuring and evaluating consumer satisfaction with products/services purchased and used in the Chinese market [26, 27]. The CCSI draws on mature international customer satisfaction index models and has been localized based on the characteristics of the Chinese market. The key variables involve customer expectations, perceived quality, perceived value, customer satisfaction, customer complaints, and customer loyalty, as shown in Fig. 4.

In Fig. 4, customer expectations affect their evaluation on perceived quality, and both customer expectations and perceived quality jointly affect perceived value. Customer expectations, perceived quality, and perceived value are the three main antecedent variables that affect the satisfaction. Customer satisfaction directly affects customer

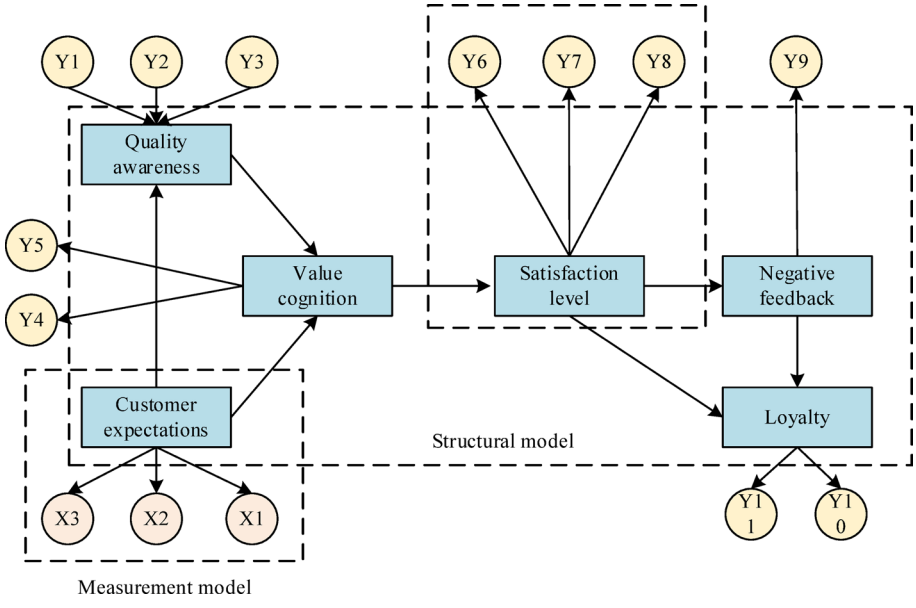


Fig. 4 Chinese customer satisfaction index model

complaints and customer loyalty, and customer complaints can also have a negative impact on customer loyalty. Based on the CCSI and the characteristics of the hotel industry, the latent variables set for research include brand image, customers' overall impression of hotel brand reputation, and market positioning. Customer expectations are the expectations for hotel services, facilities, environment, etc. before check-in. Quality perception evaluates the actual service quality of a hotel after check-in. Value perception is a subjective judgment of "cost-effectiveness" based-price and experience. Customer satisfaction exhibits the overall satisfaction. Customer complaints refer to complaints that arise from dissatisfaction. Customer loyalty refers to the willingness to repeat purchases or recommendations [28]. The constructed hotel customer satisfaction index model is presented in Fig. 5.

Based on the model, hypotheses are made regarding the relationship among brand image, customer expectations, quality perception, value perception, customer satisfaction, and customer complaints. The relationship between brand image and related variables is as follows:

H1 Brand image actively affects customer expectations;

H2 Brand image actively affects quality perception;

H3 Brand image actively affects customer satisfaction;

H4 Brand image actively affects customer loyalty.

The relationship between customer expectations and related variables is as follows:

H5 Customer expectations actively affect quality perception;

H6 Customer expectations actively affect perceived value;

H7 Customer expectations actively affect customer satisfaction.

The relationship between quality perception and related variables is as follows:

H8 Quality perception positively affects value perception;

H9 Quality perception actively affects customer satisfaction.

The relationship between perceived value and related variables is as follows:

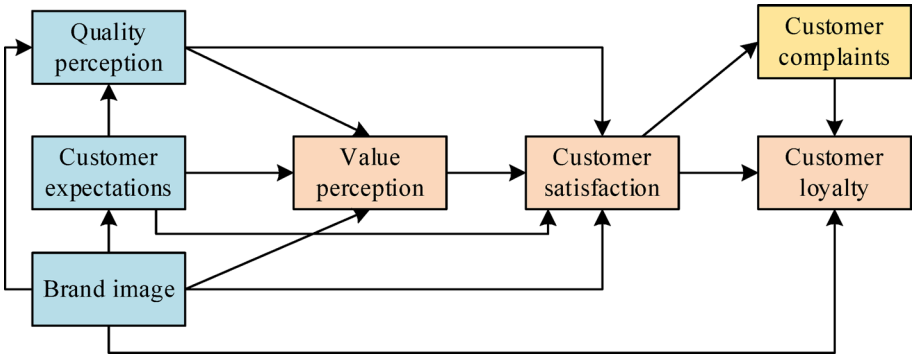


Fig. 5 Hotel customer satisfaction index model

H10 Value perception has a positive impact on customer satisfaction.

The relationship between customer satisfaction and related variables is as follows:

The relationship between customer complaints and related variables is as follows:

H11 Customer satisfaction negatively affects customer complaints;

H12 Customer satisfaction actively affects customer loyalty.

H13 Customer complaints negatively affect customer loyalty.

The constructed model reflects seven latent variables of customer satisfaction. Corresponding observable and measurable significant variables are designed for these seven latent variables. Table 1 displays the constructed customer satisfaction evaluation index.

SEM is often used to discuss the relationship between latent variables and explicit variables. Structural equation models include structural models and measurement models, where structural models mainly discuss the relationship between latent variables, as displayed in Eq. (4).

$$\eta = B\eta + \Gamma\xi + \zeta \tag{4}$$

In Eq. (4), η signifies a vector composed of endogenous latent variables. B signifies the path coefficient matrix between endogenous latent variables. Γ signifies the path coefficient matrix of exogenous latent variables to endogenous latent variables. ξ signifies a vector composed of exogenous latent variables. ζ represents the vector composed of residual terms. The measurement model mainly discusses the relationship between latent variables and explicit variables, as shown in Eq. (5).

$$\begin{cases} y = \Lambda_y \eta + \varepsilon_y \\ x = \Lambda_x \xi + \varepsilon_x \end{cases} \tag{5}$$

In Eq. (5), y signifies the observation index of η. x signifies the observation index of ξ. Λ_y and Λ_x represent the load matrix. ε_y and ε_x represent residual vectors. The modeling

Table 1 Hotel customer satisfaction evaluation index system

Structural variables (latent variables)	Measurement variable (significant variable)	Structural variables (latent variables)	Measurement variable (significant variable)
Hotel image	Overall brand impression	Quality perception	Room cleanliness
	Facility and equipment level		Comfort level of bedding
	Service quality reputation		Facility completeness
	Environmental Atmosphere Design		Bathroom experience
Value perception	Price and experience matching degree		Front desk response efficiency
	Cost effectiveness perception		Timeliness of room service
	Additional service value		Food and beverage quality
	Satisfaction with member benefits		Problem solving ability
Customer satisfaction	The gap with the ideal hotel		Comfort level in public areas
	The gap from expectations		Sound insulation effect
	Overall satisfaction		Security and Privacy
Customer complaints	Complaint frequency	Customer loyalty	Willingness to stay again
	Informal Complaints		Recommendation intention
Customer expectations	Expectations for hardware facilities		Brand Priority Selection
	Expectations for personalized services	/	/

techniques of SEM include linear structural relationship method and PLS method, among which the linear structural relationship method is based on covariance structure to construct SEM. In ideal conditions, the modeling effect of this method is better. PLS increases the minimum value of residual structure on the basis of variance, which has better application effect in causal prediction. Therefore, PLS is used for SEM. When using PLS modeling technology, the measurement model has a reflection model and a composition model, where the former is consistent with Eq. (5) and the composition model is shown in Eq. (6).

$$\begin{cases} \eta = \pi_\eta y + \delta_\eta \\ x = \pi_\xi x + \delta_\xi \end{cases} \tag{6}$$

In Eq. (6), π_η and π_ξ represent the coefficient matrix. δ_η and δ_ξ represent the residual vectors that make up the model. After using PLS technology to construct SEM, the weights of customer satisfaction evaluation indicators can be obtained to calculate customer satisfaction, as shown in Eq. (7).

$$\eta = \sum_{i=1}^{k} \omega_i y_i \tag{7}$$

In Eq. (7), ω_i signifies the weight of the indicator. k signifies the number of indicators corresponding to the latent variable. The customer satisfaction is shown in Eq. (8).

$$SC = \frac{E[\eta] - \min[\eta]}{\max[\eta] - \min[\eta]} \tag{8}$$

In Eq. (8), SC represents customer satisfaction. $E[\eta]$ represents the mean latent variable of customer satisfaction. $\min[\eta]$ and $\max[\eta]$ represent the minimum and maximum latent variables of customer satisfaction, respectively. To rigorously assess the PLS-SEM model, the evaluation includes: (a) internal consistency via Cronbach's α and composite reliability (CR, threshold > 0.70); (b) convergent validity via average variance extracted (AVE, >0.50); (c) discriminant validity via HTMT ratio (< 0.85); (d) predictive relevance via Stone–Geisser Q^2 (positive values via blindfolding); and (e) path significance via bootstrapping (5,000 resamples) with 95% bias-corrected confidence intervals. After completing the PLS-SEM model construction and parameter estimation, it is necessary to evaluate the goodness of fit of the model and the strength of the influence between variables. This study mainly reports and explains the following three key indicators: (1) R^2: represents the proportion of variance of endogenous latent variables that can be explained by other latent variables in the model, with values ranging from 0 to 1. A larger value indicates a stronger predictive ability of the model. (2) F^2: evaluates the strength of the influence of a specific antecedent variable on endogenous latent variables. F^2 values greater than 0.02, 0.15, and 0.35 respectively indicate weak, moderate, and strong effects. (3) H^2: evaluates the discriminant validity between latent variables and measures whether the correlation between measurement items of different latent variables is too high. Its value is the geometric mean of the correlation coefficients between all measurement items of two latent variables. It is generally believed that HTMT values below 0.85 or 0.90 indicate good discriminant validity. These indicators will provide a quantitative basis for subsequent model fitting analysis and path effect discussion.

4 Results

4.1 Hotel review text emotion analysis verification

The research experiment is conducted on a Linux server and the pre-trained language model is fine-tuned using the PyTorch 1.12 framework. Emotion analysis adopts the BERT Chinese basic model, where the input text is segmented and truncated to 128 characters, and the embedding layer integrates Token, Segment, and Position vectors. During the fine-tuning phase, the Adam optimizer is used to train for 10 iteration cycles, with the training, validation, and test sets divided into 7:2:1 parts. RF feature extraction integrates Word2Vec word vector, TF-IDF top five word mean, and domain emotion dictionary score, with a tree size of 200 and node splitting criterion of Gini coefficient. PLS-SEM is implemented through SmartPLS 4.0, using the Path Weighting Scheme algorithm with a maximum iteration count of 300 and a convergence threshold of 10^{-7}. The experiment takes a self-built dataset, which includes relevant reviews from 32 hotels on three different platforms. To verify the effectiveness, different features are combined separately, and the accuracy under different feature combinations is analyzed. The results are shown in Fig. 6.

Figure 6a shows the impact of domain emotion dictionary features, BERT features, and Word2Vec + BERT features on accuracy. Figure 6b shows the impact of TF-IDF + BERT, TF-IDF + Word2Vec, and all feature combinations on model accuracy. When combining domain emotion lexicon features with BERT features, the model accuracy was approximately 86%. When using only BERT features, the accuracy was about 84%. The accuracy of Word2Vec+BERT was about 88%. This indicates that combining BERT features with other features can improve the analysis accuracy, with the Word2Vec+BERT having a relatively better effect. The accuracy of the TF-IDF+BERT was about 85%. The accuracy of TF-IDF+Word2Vec was about 87%. When all features were combined, the model accuracy reached about 90%. This further demonstrates that the fused multiple features can more comprehensively capture emotional information in text, thereby significantly improving the performance of the model. The combination of all features performs best in this comparison group. The study further compares the accuracy and loss convergence during model training, and the results are shown in Fig. 7.

Figure 7a shows the training accuracy, and Fig. 7b compares the convergence loss. The accuracy showed a gradually increasing trend. Starting from the initial stage, the accuracy was about 75%. As the number of training iterations increased, the accuracy continued to improve. After a certain number of training sessions, the accuracy gradually

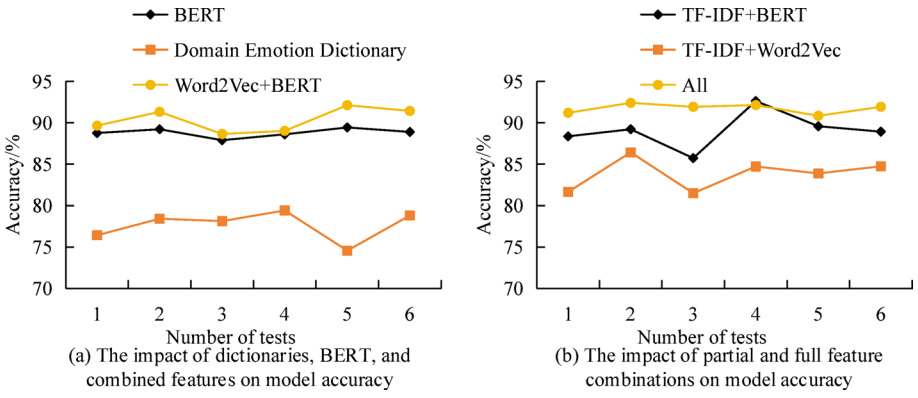


Fig. 6 The influence of different combination features on accuracy

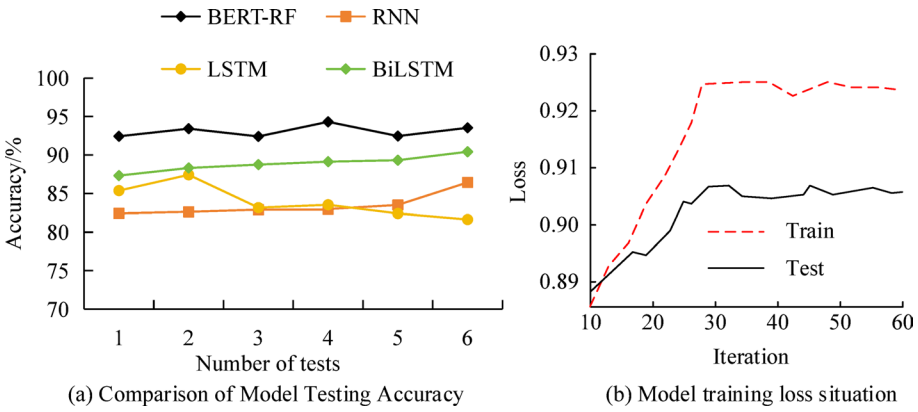


Fig. 7 Changes in accuracy and loss during model training

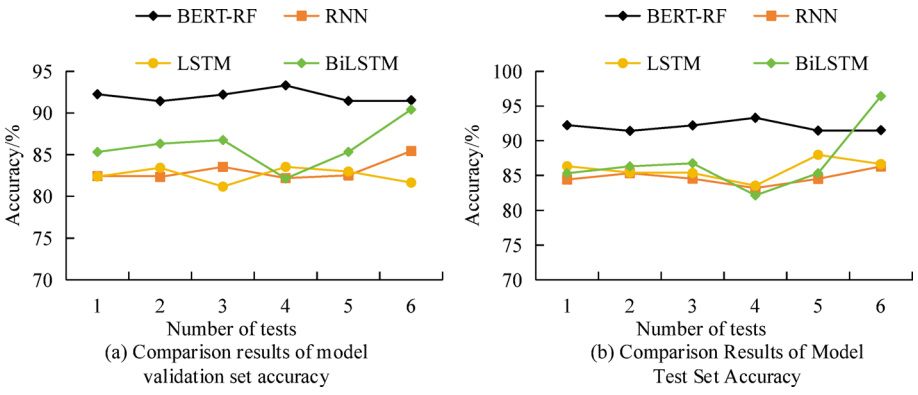


Fig. 8 Accuracy changes of different models

stabilized and eventually reached 92%. The model can learn the features in the data during the training process, and its ability to classify hotel review emotions gradually increases and stabilizes. The loss degree was relatively high in the early stages of training, around 0.6, and rapidly decreased as the training progressed. During the training process, the loss continued to decrease and eventually stabilized at 0.15. This indicates that the model can continuously optimize parameters during the training process, reduce the error between predicted values and actual values, and gradually accurately model hotel review emotion analysis. The study further analyzes the accuracy changes of different models on the validation and test sets. The results are shown in Fig. 8.

Figure 8a presents the accuracy changes on the validation set, and Fig. 8b presents the accuracy changes on the test set. The accuracy of the BERT-RF on the validation set was about 89%, and the model exhibited high accuracy on the validation set, indicating that it also has strong emotion classification ability on untrained data. The accuracy of the RNN on the validation set was about 78%, which was slightly lower than that of the BERT-RF model, indicating that the generalization ability of the RNN model in handling hotel review emotion analysis tasks was relatively weak. Overall, the BERT-RF model shows significant advantages in both the validation and test sets, with significantly higher accuracy than that of the RNN and LSTM models, and stable performance with strong generalization ability on both datasets. This fully demonstrates that the BERT-RF model has better performance and reliability in hotel review emotion analysis tasks, and can more accurately classify and analyze the emotions of hotel users' online reviews,

providing strong model support for subsequent research on improving hotel management based on emotion analysis. Further analysis is conducted on the precision, F1 score, and recall using the designed model and other models, as presented in Table 2. To rigorously evaluate the proposed BERT-RF, the original BERT-base-Chinese is included as a direct baseline, which shares the same backbone and training setup, thereby isolating the contribution of the RF-based feature enhancement.

According to Table 2, the BERT-RF model performed the best in hotel review emotion classification tasks, with significantly higher precision, F1 score, and recall rate than those of RNN and LSTM models in classifying negative, neutral, and positive reviews. This fully demonstrates that the BERT-RF model has stronger advantages in handling emotional analysis of hotel users' online reviews, and can more accurately identify and classify reviews with different emotional tendencies, providing a more reliable basis for hotels to improve management based on emotional analysis. However, the performance of RNN and LSTM models in this task is relatively poor, which may have limitations when dealing with complex text emotion features. Compared with the original BERT, BERT-RF improves overall accuracy from 88.0% to 90.1%, with a McNemar test chi-square of 14.73 ($p < 0.001$), confirming that the statistical feature fusion yields statistically significant gains, especially in neutral and negative reviews.

To quantify the individual contribution of each feature module and validate the effectiveness of the dynamic re-weighting fusion mechanism over simple feature concatenation, a systematic ablation experiment is conducted. Five model variants are compared on the same balanced test set:

Model A (BERT baseline): Only the frozen 768-dim BERT [CLS] embedding is used as input to a standard RF classifier (no statistical features).

Model B (BERT + RF with concatenation): The 768-dim BERT embedding is concatenated with all statistical features (Word2Vec, TF-IDF, lexicon) and fed into RF. This represents the naive stacking baseline.

Model C (BERT + W2V + TF-IDF, no lexicon): Same as Model B, but excluding the domain lexicon score.

Model D (BERT + lexicon, no W2V/TF-IDF): Same as Model B, but excluding Word2Vec and TF-IDF.

Model E (Full model with dynamic re-weighting): the proposed method, where RF internally adjusts feature sampling weights based on Gini impurity gain comparison during training.

Table 2 The precision, F1 score, and recall of emotion classification for different models

Model	Emotion classification	Precision	F1 score	Recall
BERT	−1	0.78	0.79	0.80
	0	0.79	0.74	0.69
	1	0.87	0.88	0.90
BERT-RF	−1	0.81	0.82	0.83
	0	0.82	0.77	0.72
	1	0.90	0.91	0.93
RNN	−1	0.68	0.70	0.73
	0	0.72	0.68	0.64
	1	0.83	0.87	0.91
LSTM	−1	0.66	0.69	0.73
	0	0.71	0.66	0.61
	1	0.84	0.84	0.84

Table 3 Results of ablation test

Model variant	Accuracy	Macro F1	F1 (Positive)	F1 (Neutral)	F1 (Negative)
A. BERT only (RF baseline)	0.842	0.812	0.862	0.742	0.832
B. BERT + all stats (concatenation)	0.887	0.865	0.898	0.788	0.862
C. BERT + W2V + TF-IDF (no lexicon)	0.871	0.848	0.885	0.765	0.847
D. BERT + lexicon (no W2V/TF-IDF)	0.863	0.839	0.877	0.758	0.841
E. Full model (dynamic re-weighting)	0.901	0.882	0.91	0.81	0.88

Table 4 Fitting analysis of customer satisfaction model

Variable	H^2	R^2	F^2	Cronbach's α	CR	AVE	Highest HTMT	Q^2
Brand image	0.683	/	/	0.813	0.872	0.621	0.341	/
Customer expectations	0.648	0.692	0.482	0.784	0.851	0.584	0.483	0.532
Quality perception	0.672	0.509	0.351	0.836	0.887	0.638	0.524	0.418
Value perception	0.581	0.612	0.391	0.862	0.908	0.712	0.724	0.501
Customer satisfaction	0.759	0.713	0.542	0.801	0.859	0.593	0.724	0.586
Customer complaints	0.492	0.549	0.284	0.761	0.824	0.552	0.643	0.347
Customer loyalty	0.581	0.602	0.313	0.829	0.883	0.651	0.698	0.427
Whole	0.631	0.621	0.392	/	/	/	/	/
Thresholds	/	/	/	> 0.70	> 0.70	> 0.50	< 0.85	> 0

Table 3 presents the accuracy, macro F1, and per-class F1 scores for all variants.

The ablation results reveal three key insights. First, comparing Model A vs. Model B, adding statistical features improves accuracy by 4.5%, demonstrating that BERT's contextual embeddings alone are insufficient for capturing domain-specific intensity and keyword frequency patterns in short hotel reviews. Second, comparing Models C and D against Model B, both the lexicon score and the TF-IDF/Word2Vec features contribute positively, but the lexicon score has a slightly stronger effect on negative review identification (F1 increases from 0.847 to 0.862 when adding the lexicon), confirming its value for detecting subtle complaint expressions. Most importantly, comparing Model B (simple concatenation) with Model E (proposed dynamic re-weighting), the latter achieves a 1.4% absolute accuracy gain and a 0.017 gain in macro F1, even though both use the same set of features. This improvement is attributed to the adaptive weighting mechanism, which prevents redundant statistical features from overshadowing BERT embeddings when they carry little marginal information, while amplifying those that complement semantic understanding. This confirms that the fusion strategy, rather than mere feature accumulation, constitutes the substantive algorithmic improvement of the proposed BERT-RF framework.

4.2 Verification of hotel customer satisfaction analysis

The study first conducts a fitting analysis on the customer satisfaction model, as presented in Table 4. The H^2 of customer expectations was 0.648, with an R^2 value of 0.692 and an F^2 value of 0.482. The high values of H^2 and R^2 indicate that the customer expectation can be well explained by other variables, and their own influence on other variables is significant. The F^2 value was 0.482, demonstrating a strong correlation effect between customer expectations and other variables. From the fitting analysis results, most of the variables had good fitting indicators, indicating that it could explain the relationships between these variables. However, the fitting index of customer complaints was relatively low. This may further analyze its role and influencing factors in the model,

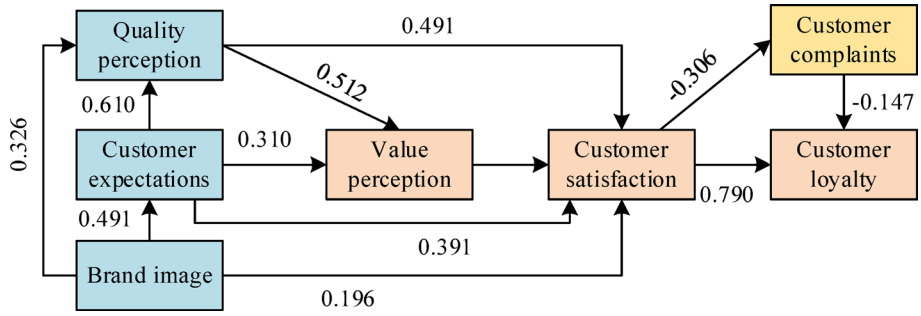


Fig. 9 Revised customer satisfaction model path coefficient test

Table 5 Latent variable correlation coefficient matrix

/	Brand image	Customer expectations	Quality perception	Value perception	Customer satisfaction	Customer complaints	Customer loyalty
Brand image	1	/	/	/	/	/	/
Customer expectations	0.041	1	/	/	/	/	/
Quality perception	0.162	0.303	1	/	/	/	/
Value perception	0.113	0.084	0.342	1	/	/	/
Customer satisfaction	0.021	0.042	0.093	0.774	1	/	/
Customer complaints	−0.121	−0.071	−0.079	−0.373	−0.352	1	/
Customer loyalty	0.082	0.094	0.102	0.702	0.749	0.327	1

or consider modifying and improving the explanatory ability for customer complaints as a variable.

Table 4 reports all key measurement and structural model evaluations in one consolidated view. All Cronbach's α and CR values exceed 0.70, all AVE values exceed 0.50, and the highest HTMT across all construct pairs is 0.724 (between Value Perception and Customer Satisfaction), well below the 0.85 threshold, collectively confirming strong reliability, convergent validity, and discriminant validity. The positive Q^2 values (ranging from 0.347 to 0.586) confirm satisfactory predictive relevance for all endogenous variables. The R^2 and F^2 values, as previously interpreted, further indicate adequate explanatory power and effect sizes. Bootstrapping (5,000 resamples) confirms that 12 out of 13 hypothesized paths are statistically significant at $p < 0.05$. The only non-significant path is H4 (Brand Image → Customer Loyalty, β = 0.068, $p = 0.073$), which is consequently excluded from the final revised model. All significant path coefficients and their bias-corrected 95% confidence intervals are consistent with the directions and magnitudes reported in Fig. 9.

After correcting the model path coefficient test, the correlation coefficient test is conducted on the latent and significant variables. The correlation coefficient matrix of the latent variables is presented in Table 5. The correlation coefficient between value perception and customer satisfaction was 0.774, which was the strongest positive correlation in the table. This indicates that if customers have a high subjective judgment for "cost-effectiveness", their overall satisfaction is also higher, which verifies H10. The correlation

coefficient between customer satisfaction and customer loyalty was 0.749, showing a strong positive correlation. High customer satisfaction may result in repeat purchases or recommendation behavior, consistent with H12. The correlation coefficient between value perception and customer loyalty was 0.702, signifying a strong correlation. This indicates that value perception directly affects satisfaction, and indirectly affects loyalty through satisfaction, reflecting the transmission path of "cost-effectiveness → satisfaction → loyalty". The correlation coefficient was 0.342, showing a moderate positive correlation. If customers have a high actual evaluation of hotel service quality, they are more likely to recognize the value of the hotel, supporting H8. The correlation coefficient between quality perception and customer expectations was 0.303, indicating a positive correlation. This may reflect the match between customers' expectations before check-in and their actual quality perception. Customers with higher expectations may be more sensitive to quality, but the correlation coefficient is lower, indicating limited correlation between the two. The correlation coefficient between customer complaints and value perception was − 0.373, and the correlation coefficient with customer satisfaction was − 0.352, both showing a moderate negative correlation. This indicates that the lower the perceived value and satisfaction, the more likely customers are to engage in complaining behavior, verifying H11. The correlation coefficient between customer complaints and customer loyalty is 0.327, appearing positive at the bivariate level. However, this zero-order correlation should not be interpreted as evidence that complaints enhance loyalty. Instead, it reflects a classic statistical suppression effect. In the structural model, after controlling for Customer Satisfaction and Value Perception, which mediate the complaint-loyalty relationship, the direct path coefficient for H13 is significantly negative ($\beta = -0.242$, $p < 0.001$), consistent with theoretical expectations. The positive raw correlation is plausibly attributable to the nature of the sample: customers who are frequent reviewers on OTA platforms (e.g., Ctrip, Meituan) tend to be both more loyal (higher revisit rates) and more vocal about service failures, thereby creating a positive confounding at the descriptive level. Thus, caution is advised against equating complaint frequency with complaint resolution quality. The structural evidence supports that unaddressed complaints ultimately undermine loyalty, whereas the raw correlation captures a composition effect of active users rather than a causal benefit of complaints.

The study further analyzes the customer satisfaction index, as displayed in Table 6. Customers have extremely high expectations for hotel hardware, which may be a key focus of hotel promotion or a general trend in the industry. It is necessary to ensure that the actual facilities meet the standards to avoid a gap in expectations. Room cleaning is a fundamental aspect of hotel services. A high score indicates that this dimension has met the basic needs and can be maintained as a benchmark for service. The overall satisfaction score was relatively high, indicating that the hotel had not made significant mistakes in core services and the customer experience was still acceptable. The willingness to stay again and the priority of brand selection reflect the customer's recognition, which is the core manifestation of loyalty. This advantage needs to be consolidated through continuous optimization. The score for the gap between the ideal hotel and expectations was lower than the overall satisfaction, indicating that the customer's actual experience did not fully meet their psychological expectations. It is necessary to focus on narrowing the "expectation-experience" gap. There was a significant gap between customer expectations and actual quality perception and value perception. Excessive promises should

Table 6 Satisfaction index of latent and significant variables in customer satisfaction

Structural variables (latent variables)	Index	Measurement variable (significant variable)	Index	Structural variables (latent variables)	Index	Measurement variable (significant variable)	Index
Hotel image	6.242	Overall brand impression	6.561	Quality perception	6.353	Room cleanliness	6.790
		Facility and equipment level	6.143			Comfort level of bedding	6.132
		Service quality reputation	6.542			Facility completeness	6.138
		Environmental Atmosphere Design	6.023			Bathroom experience	6.123
Value perception	6.325	Price and experience matching degree	6.543			Front desk response efficiency	6.132
		Cost effectiveness perception	6.261			Timeliness of room service	6.231
		Additional service value	6.352			Food and beverage quality	6.325
		Satisfaction with member benefits	6.143			Problem solving ability	6.152
Customer satisfaction	6.389	The gap with the ideal hotel	6.134			Comfort level in public areas	6.323
		The gap from expectations	6.210			Sound insulation effect	6.151
		Overall satisfaction	6.523			Security and Privacy	6.173
Customer complaints	5.670	Complaint frequency		Customer loyalty	6.170	Willingness to stay again	6.230
		Informal Complaints				Recommendation intention	6.110
Customer expectations	7.447	Expectations for hardware facilities	7.832			Brand Priority Selection	6.235
		Expectations for personalized services	7.062	/		/	

be avoided and the actual service experience can be improved, especially in weak areas such as front desk response and bathroom facilities. To address the low perceived value scores, customers can enhance their subjective perception for "cost-effectiveness" by optimizing the pricing system, adding free additional services, and improving member benefits.

The study further analyzes the correlation between emotion classification, satisfaction drive, and management shortcomings, as shown in Table 7. The overall accuracy of emotion classification for all comments reached 90%, and the p-value was below 0.001, demonstrating that the result had extremely high statistical significance, indicating that the model performed well in the overall emotion classification task. The value perception → satisfaction path coefficient was 0.774, with a p-value below 0.001, demonstrating that value perception has positive direct impacts on satisfaction. The higher the subjective judgment on the "cost-effectiveness", the higher their overall satisfaction. The customer's expected value for hardware facilities was 7.832, but the actual perceived value was only 6.325. This reflects a significant gap between the customer's high expectations for hardware facilities and the actual perceived value, which is one of the important reasons for the low perceived value score. The room cleanliness score of 6.790 was relatively high, indicating that the hotel performed better in this area and met the basic needs of

Table 7 Analysis results of the correlation between emotion classification, satisfaction drive, and management shortcomings

Dimension	Indicators/Variables	Classification/Grouping	Value	Significance
Emotion classification performance	Overall accuracy	All comments	90%	$p < 0.001$
	Positive review recall rate	Emotional label = 1	93%	$p < 0.001$
	Accuracy rate of negative comments	Emotional label = − 1	81%	$p < 0.001$
Satisfaction path coefficient	Value perception → Satisfaction	Direct effect	0.774	$p < 0.001$
	Customer Expectations → Satisfaction	Direct effect	− 0.21	$p < 0.001$
	Customer complaints → Value perception	Direct effect	− 0.373	$p < 0.001$
	Satisfaction → Loyalty	Direct effect	0.749	$p < 0.001$
Key significant variable score	Value perception related	Expectations for hardware facilities	7.832	/
		Actual value perception	6.325	/
	Quality perception related	Room cleaning	6.790	/
		Bathroom experience	6.123	/
	Complaint handling related	Front desk response efficiency	6.132	/
	Loyalty related	Willingness to stay again	6.230	/

customers. The bathroom experience score of 6.123 was relatively low, which was a weak link in quality perception and needed to be improved as a key focus.

5 Discussion

The study systematically analyzes the emotional tendencies and satisfaction driving mechanisms of hotel users' online reviews by integrating multi-feature text analysis with an improved PLS-SEM model. The research results validate the effectiveness of the BERT-RF in emotion analysis, and reveal the key role of value perception as the core driver of satisfaction through PLS-SEM path analysis, as well as the significant negative impact of customer complaints on value perception.

Existing scholars often rely on single text analysis, while the research method integrates text features, statistical features, and domain emotion dictionaries, significantly improving classification accuracy. This is consistent with the conclusion drawn by El-Said O et al. to emphasize the complementarity of multi-source data [29]. Further research quantified the gain mechanism of multi-modal fusion through feature combination experiments, and all feature combinations improved the accuracy from 84 to 90%, verifying the warning about the "risk of generalization from a single data source" drawn by Wu [30]. Compared to traditional models, BERT-RF performs better in negative comment recognition and neutral comment classification. This result supports the improvement approach to the insufficient utilization of contextual semantics drawn by Rassal et al. [31]. The study further addresses the limitations of BERT in handling sparse features by introducing the feature extraction capability of RF, providing a new solution for fine-grained emotion analysis. Path analysis shows that value perception has the strongest direct impact on satisfaction, far exceeding quality perception and customer expectations. This discovery echoes the doubts drawn by Rafiki et al. [32]. Traditional service quality scales may underestimate the weight of "cost-effectiveness" in consumer decision-making. Based on explicit variable measurement, the significant gap between customers' expectations of "hardware facilities" and their actual "value perception" is a

shortcoming in satisfaction, providing empirical evidence for hotels to optimize pricing strategies and additional services.

By accurately identifying weak links through satisfaction indices, hotels can prioritize allocating resources to areas related to "cost-effectiveness perception" rather than blindly improving all service dimensions. Based on the empirical findings, the study propose four actionable managerial strategies:

(1) Pinpoint service defects: Focus physical upgrades and staff training on the weakest links identified by sentiment analysis: bathroom experience (6.123) and front-desk response (6.132).
(2) Value-driven pricing: Given value perception's strongest impact on satisfaction (β = 0.774) and the large expectation-perception gap (7.832 vs. 6.325), adopt transparent pricing with added-value bundles (e.g., late check-out, loyalty points) to boost perceived cost-effectiveness without lowering base rates.
(3) Review triage: Use BERT-RF to classify reviews by sentiment and topic: prioritize high-intensity negative reviews for immediate response and aggregate recurring low-score themes for periodic improvement planning.
(4) Proactive complaint interception: Since complaints reduce value perception (β = −0.373), set a daily sentiment threshold to trigger alerts, enabling managers to proactively contact dissatisfied guests before negative reviews are posted.

6 Conclusion

The research aims to construct a closed-loop system of "emotion analysis—problem diagnosis—management optimization", which accurately analyzes the emotion of hotel text reviews and quantifies the satisfaction driving mechanism, helping hotels shift from passive response to negative reviews to active optimization, and improving customer trust and platform ratings. The research employed a BERT-RF model that combines BERT's semantic encoding with RF's feature integration capability for emotion analysis of hotel reviews. Based on the CCSI, a hotel specific indicator system was designed with seven latent variables. The improved PLS-SEM method was used to analyze path relationships, and the satisfaction index was calculated using explicit variable weights. The emotion classification accuracy of the BERT-RF reached 90%, with a positive review recall rate of 93% and a neutral review accuracy rate of 82%. The customer had high expectations for hardware facilities, with a score of 7.832, but the lowest perceived value score was only 6.325, and the "expectation-experience gap" path coefficient significantly lowered the overall satisfaction. The research has verified that multi-modal data fusion improves emotion analysis accuracy, and the BERT-RF model serves as an effective approach for hotel review classification. Hotels need to prioritize optimizing their perceived "cost-effectiveness" and narrowing the gap between "expectation-experience". The PLS-SEM model does not consider the potential impact of moderating variables like cultural differences and regional characteristics on satisfaction. Future research will introduce time series analysis to track the long-term evolution of satisfaction driving factors and verify the differences in the mechanism of negative evaluation generation in cross-cultural scenarios.

Author contributions
X.X. provided the concept, acquired the funding and wrote the draft, and C.Q. prepared the figures and tables, Y.Q. revised the paper critically. All authors reviewed this paper carefully and approved this submission.

Funding
The research is supported by: 2025 Hainan Provincial Department of Education Higher Education Research Project: Research on the Impact of Corporate Social Responsibility of Tourism Hotel Enterprises Hainan Province on Green Consumption Behavior Based on S-O-R Theory (Fund Number: Hnky2025-97).

Data availability
All data generated or analyzed during this study are included in this article. Further enquiries can be directed to the corresponding author.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 19 May 2026 / Accepted: 3 September 2026

References

1. Gao J. Optimizing hotel revenue management through dynamic pricing algorithms and data analysis. J Comput Methods Sci Eng. 2025;25(2):1200–9. https://doi.org/10.1177/14727978241298467.
2. Moreno Brito YL, Ban HJ, Kim HS. Ecological hotels' customer satisfaction through text mining of online reviews: a case of Ecuador hotels. J Hosp Tour Insights. 2024;7(3):1532–52. https://doi.org/10.1108/JHTI-09-2022-0467.
3. Nicolau JL, Xiang Z, Wang D. Daily online review sentiment and hotel performance. Int J Contemp Hosp Manag. 2024;36(3):790–811. https://doi.org/10.1108/IJCHM-05-2022-0594.
4. Saha P, Biswas A. Seal the deal: unleashing the magnitude of online reviews, website quality and trust for seamless hotel reservations. Int J Prod Perform Manag. 2025;74(4):1125–55. https://doi.org/10.1108/IJPPM-04-2024-0272.
5. Akel G, Noyan E. Exploring the criteria for a green and smart hotel: insights from hotel managers' perspectives. J Hosp Tour Insights. 2024;7(5):2992–3012. https://doi.org/10.1108/JHTI-08-2023-0555.
6. Lee HSK, Wang J, Moreno-Brito YL, et al. Linguistic insights into customer satisfaction: an exploratory analysis of online reviews for gaming destination resorts in Las Vegas. J Hosp Tour Technol. 2025;16(1):73–90. https://doi.org/10.1108/JHTT-05-2023-0147.
7. Hidayati NN. Improving aspect-based sentiment analysis for hotel reviews with latent Dirichlet allocation and machine learning algorithms. Regist J Ilm Tek Sist Inf. 2023;9(2):144–59. https://doi.org/10.26594/register.v9i2.3441.
8. Li C, Kwok L, Xie KL, Liu J, Ye Q. Let photos speak: the effect of user-generated visual content on hotel review helpfulness. J Hosp Tour Res. 2023;47(4):665–90. https://doi.org/10.1177/10963480211019113.
9. Xu J, Zhang J, Song LY, et al. Fine-grained sentiment analysis of online reviews based on ROBERTa-BiLSTM-CRF. Syst Eng Theory Pract. 2023;43(12):3519–36. https://doi.org/10.12011/SETP2022-2001.
10. Xun J, Wang Y, Chong WK, Wong E, Fan X. Maze of conveniences: re-examining hotel reviews and social media data. Ind Manag Data Syst. 2025;125(5):1946–76. https://doi.org/10.1108/IMDS-09-2024-0881.
11. Kim JM, Lee E, Yoon Y. Bringing culture into the picture: cross-cultural differences in online customer reviews. Int Mark Rev. 2023;40(3):528–46. https://doi.org/10.1108/IMR-07-2020-0167.
12. Wąsowicz-Zaborek E. Content analysis of hotel reviews as a quality management tool: preliminary verification of the SERVQUAL scale. Turyzm. 2023;33(1):29–40. https://doi.org/10.18778/0867-5856.33.1.03.
13. Petrescu M, Gironda J, O'Leary KB. Consumer-brand heuristics in luxury hotel reviews. J Prod Brand Manag. 2024;33(4):430–42. https://doi.org/10.1108/JPBM-01-2023-4303.
14. Lopes AI, Malthouse EC, Dens N, De P. Is webcare good for business? A study of the effect of managerial response strategies to online reviews on hotel bookings. J Serv Manag. 2024;35(6):22–41. https://doi.org/10.1108/JOSM-05-2023-0219.
15. Guo Q, Yan J. The impact of online reviews on hotel ratings through the lens of elaboration likelihood model: a text mining approach. KSII Trans Internet Inf Syst. 2023;17(10):2609–26. https://doi.org/10.3837/tiis.2023.10.001.
16. Kazakov S, Cuesta-Valiño P, Butkouskaya V, Muravsky D. Improving guest satisfaction by identifying hotel service micro-elements failures through deep learning of online reviews. Cuad Gest. 2025;25(1):71–88. https://doi.org/10.5295/cdg.242191sk.
17. Chiu W, Cho H. The model of goal-directed behavior in tourism and hospitality: a meta-analytic structural equation modeling approach. J Travel Res. 2022;61(3):637–55. https://doi.org/10.1177/0047287521991242.
18. Hasan AAT, Rahman MT. Factors influencing green hotel revisit intentions after the COVID-19 in Bangladesh. Int J Tour Cities. 2023;9(1):143–58. https://doi.org/10.1108/IJTC-03-2022-0065.
19. Shehata AE, Khan MA, Khalid R, Raza M, Selem KM. Consequences of paradoxical leadership in the hotel setting: moderating role of work environment. J Hosp Mark Manag. 2023;32(5):670–93. https://doi.org/10.1080/19368623.2023.2190318.
20. Rezaei F, Raeesi Vanani I, Jafari A, Kakavand S. Identification of influential factors and improvement of hotel online user-generated scores: a prescriptive analytics approach. J Qual Assur Hosp Tour. 2024;25(4):1070–109. https://doi.org/10.1080/1528008X.2022.2146620.
21. Nguyen VT, Phong LT, Chi NTK. The impact of AI chatbots on customer trust: an empirical investigation in the hotel industry. Consum Behav Tour Hosp. 2023;18(3):293–305. https://doi.org/10.1108/CBTH-06-2022-0131.

22. Ahmed MEAM. Development of a management framework for energy efficiency for sustainability of hotels in Jeddah, Saudi Arabia using structural equation modelling. J Adv Res Appl Sci Eng Technol. 2024;36(1):159–75. https://doi.org/10.37934/araset.36.1.159175.
23. Christin GAD, Nugraha AKN. The impact of online review and price on hotel booking intention at online travel agency: trust as a mediating variable. Int J Electron Commer Stud. 2023;13(4):159–84. https://doi.org/10.7903/ijecs.2115.
24. Elshaer AM, Marzouk AM. Memorable tourist experiences: the role of smart tourism technologies and hotel innovations. Tour Recreat Res. 2024;49(3):445–57. https://doi.org/10.1080/02508281.2022.2027203.
25. Olorunsola VO, Saydam MB, Lasisi TT. Customer experience management in capsule hotels: a content analysis of guest online review. J Hosp Tour Insights. 2023;6(5):2462–83. https://doi.org/10.1108/JHTI-03-2022-0113.
26. Yu W, Cui F, Hou Z. The evolution of consumers' demand for hotels under the public health crisis: opinion mining from online reviews. Curr Issues Tour. 2023;26(12):1974–90. https://doi.org/10.1002/csr.2508.
27. Murtaza SH, Khan A, Mustafa SM. Eco-centric success: stakeholder approaches to sustainable performance via green improvisation behavior and environmental orientation in the hotel industry. Bus Strategy Environ. 2024;33(7):7273–86. https://doi.org/10.1002/bse.3860.
28. Jalali N, Moon S, Kim MY. Profiling diverse reviewer segments using online reviews of service industries. J Mark Anal. 2023;11(2):130–48. https://doi.org/10.1057/s41270-022-00163-w.
29. El-Said O, Elhoushy S, Al Bulushi S. How do online review valence and ratings interact with consumer-generated visuals? Serv Ind J. 2024;44(15–16):1181–210. https://doi.org/10.1080/02642069.2022.2142214.
30. Wu J, Fu C, Zhao E, Sun S, Wang S. What makes a better hotel selection? A hybrid approach considering tourists' preference and review helpfulness. Curr Issues Tour. 2025;28(10):1570–94.
31. Rassal C, Correia A, Serra F. Understanding online reviews in all-inclusive hotels servicescape: a fuzzy set approach. J Qual Assur Hosp Tour. 2024;25(6):1607–34. https://doi.org/10.1080/1528008X.2023.2167761.
32. Rafiki A, Nasution MDTP, Rossanty Y. CRM impact on customer satisfaction in Islamic-based hotels: exploring a mediation model. J Islam Mark. 2024;15(12):3425–44. https://doi.org/10.1108/JIMA-08-2023-0265.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.